

Task 3: Iris Flower Classification

Project Overview

This project classifies Iris flowers into three different species using a Machine Learning classification model. The model analyzes flower measurements and predicts whether the flower belongs to the Setosa, Versicolor, or Virginica species.

Objective

Build a machine learning model to accurately classify Iris flower species based on their physical measurements.

Technologies Used

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn
- Google Colab

Workflow

- Import the required libraries.
- Load the Iris dataset.
- Preprocess and explore the data.
- Visualize the dataset.
- Split the data into training and testing sets.
- Train the machine learning model.
- Evaluate the model's performance.
- Predict the species of new Iris flowers.

Result

The model successfully classifies Iris flower species with high accuracy. Data visualizations and evaluation metrics demonstrate the effectiveness of the classification model.